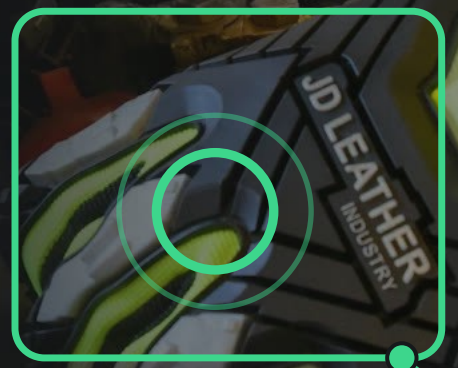
praxis

STEP 4/9

Instructions that can finally see how you're doing

An AI coach for new maintenance technicians. It watches the work through a first-person camera and says the next step.



Seat the clamp past the hose bead

CONCEPT MOCK · STOCK PHOTO

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

THE PROBLEM

A new maintenance technician needs about a year of a senior tech's time to work alone

>1 year

of on-the-job training, with a senior tech as the teacher.

$11–23k

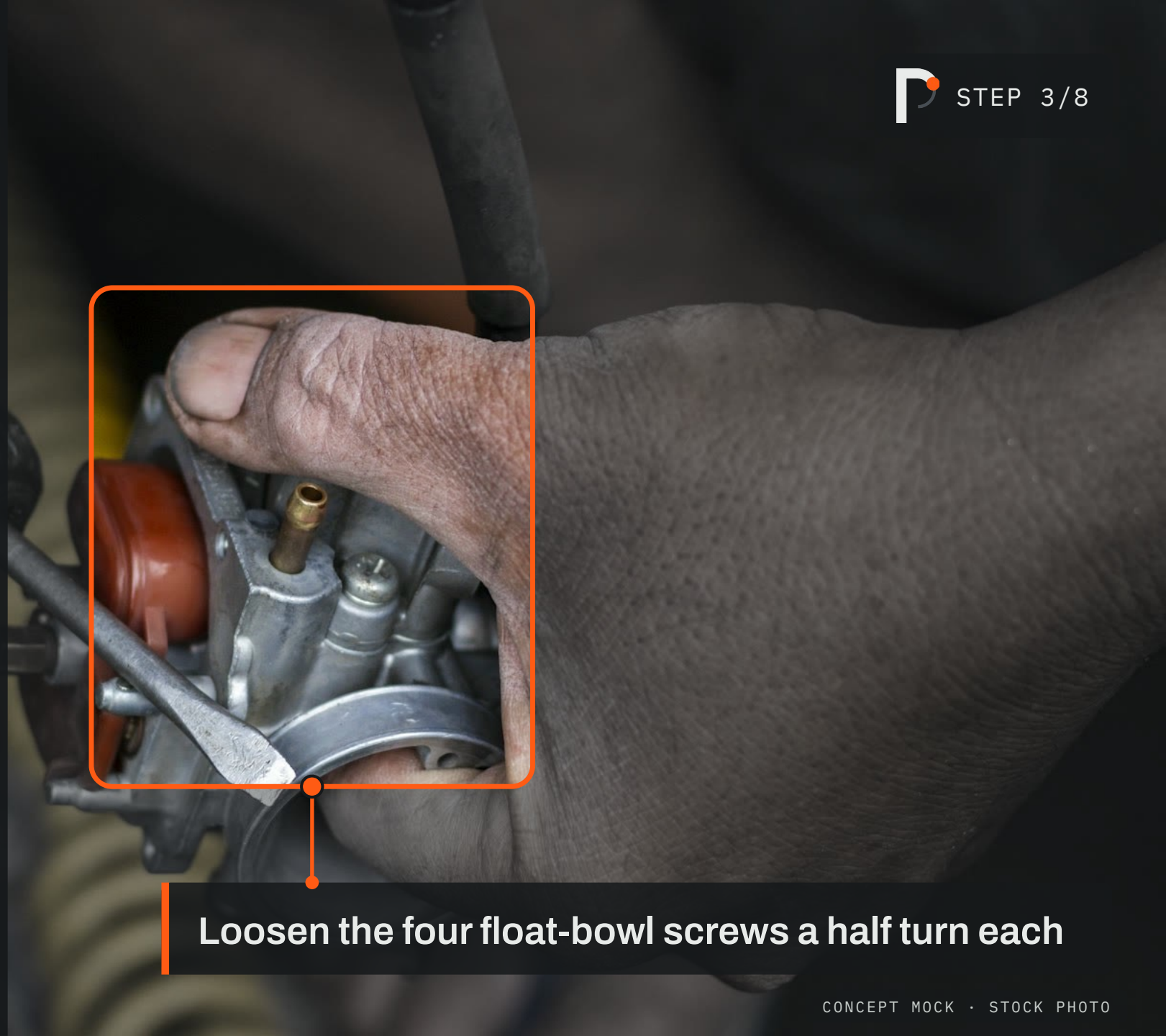
of senior-tech time per new hire (our estimate).

BLS Occupational Outlook Handbook · BLS Employer Costs for Employee Compensation, June 2025 · estimate labeled as ours

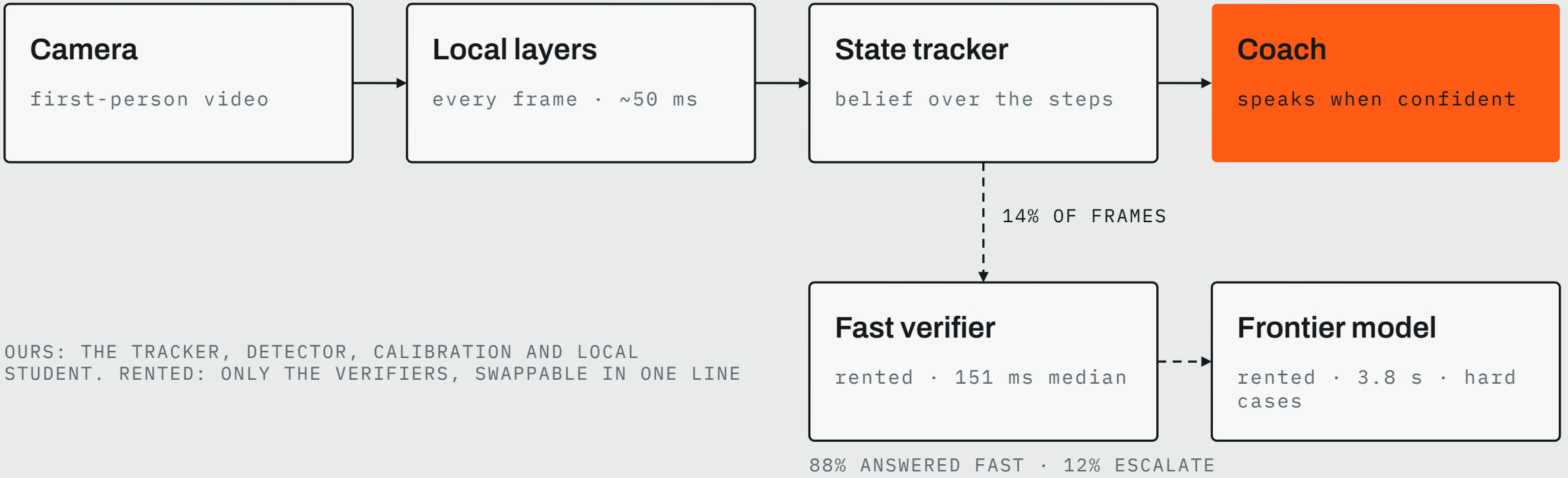
THE PRODUCT

Praxis watches the work and coaches one step at a time

LOAD	the manuals, checklists and work orders a shop already has
COACH	on a phone at the bench or on glasses, one short line at a time
VERIFY	critical steps and keep a record with keyframes

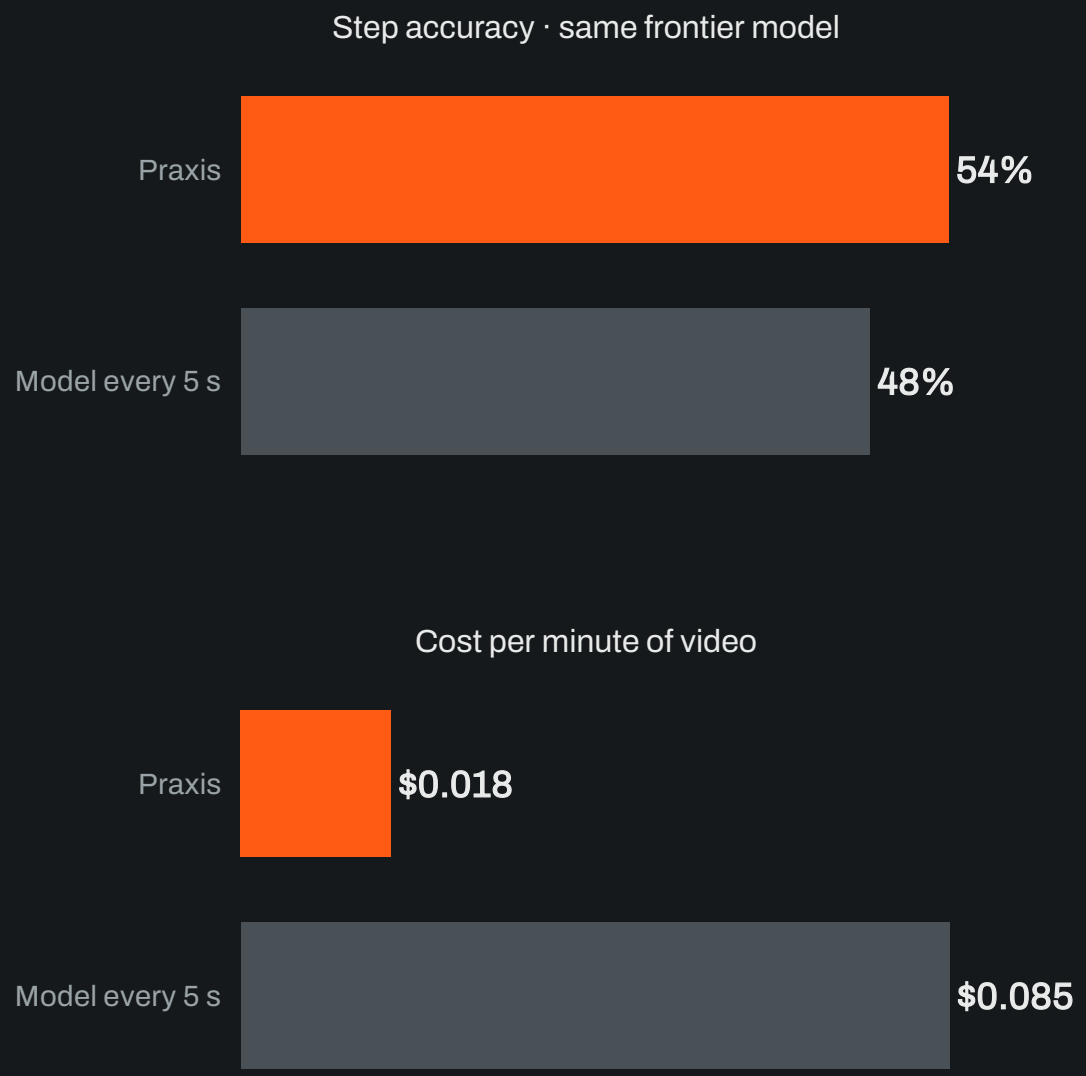


Cheap models watch every frame. The big model is called only when something happens



OURS: THE TRACKER, DETECTOR, CALIBRATION AND LOCAL STUDENT. RENTED: ONLY THE VERIFIERS, SWAPPABLE IN ONE LINE

Measured on 48 held-out first-person clips · full engine in appendix A1



EVIDENCE

Same frontier model, with our tracking: 13% more accurate at a fifth of the cost

13%

more accurate at tracking the current step

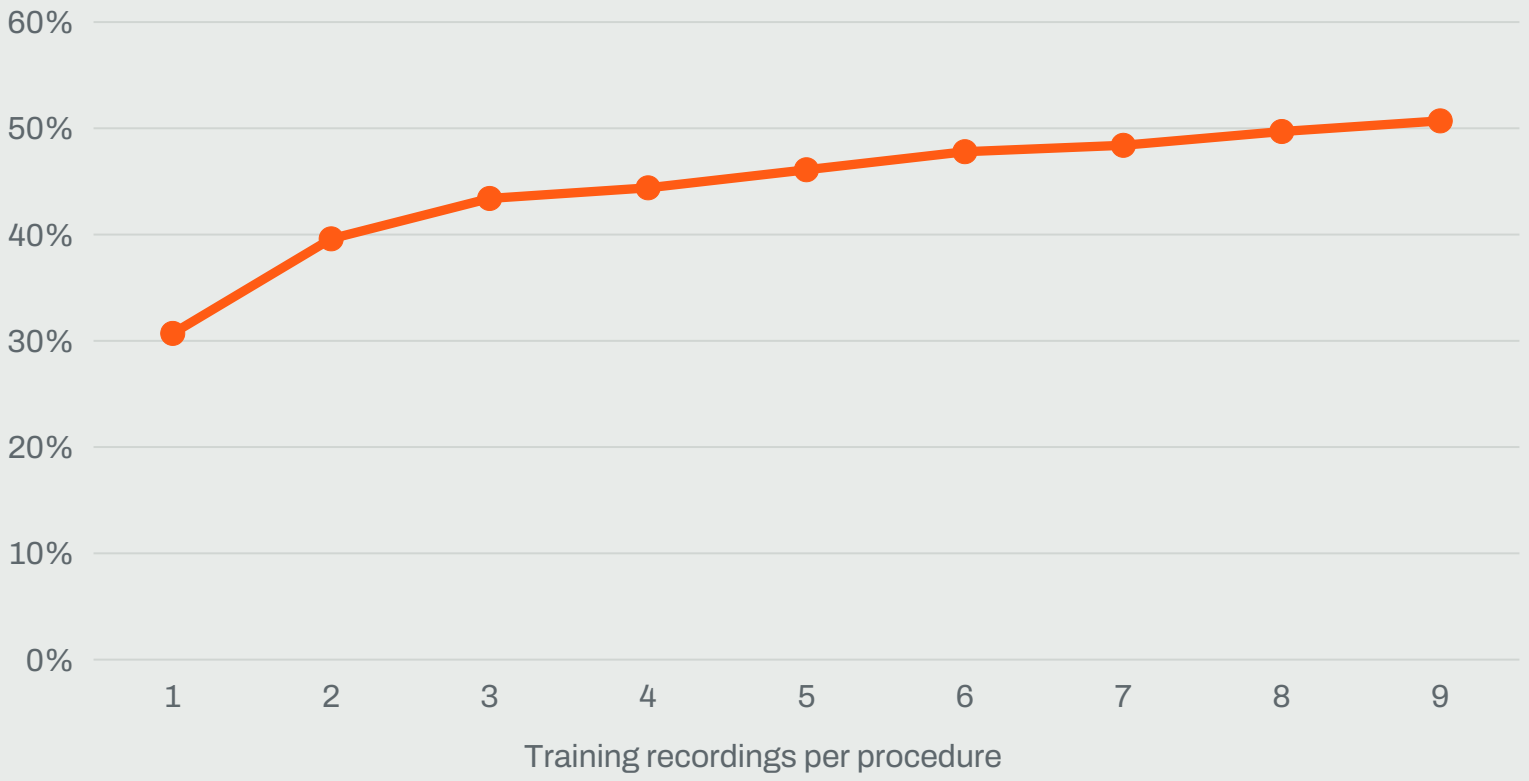
1/5

the cost per minute of video

48 held-out clips of public cooking video · one run each

IT COMPOUNDS

Our models get better with every recording, so every customer session makes Praxis better



30.7% → 50.7%

step accuracy on videos it never trained on, from 1 to 9 recordings. Still rising.

Today this is public video. Next, it's our customers' sessions: step-labeled skilled work that nobody else has.

Retrained on random subsets of public train-split recordings, 5 seeds per point

WHY NOW

This year, frontline teams are choosing what replaces their last generation of tools

28 MAY 2026
Autodesk agrees to buy MaintainX for $3.6B

16 JUN 2026
Viture + NVIDIA announce coaching glasses

SEPT 2026
Meta opens its glasses' camera to apps

31 DEC 2026
Microsoft ends Dynamics 365 Guides

Q1 2027
Helix and Android XR display glasses ship

And it finally costs out: about $1 an hour of video for us, vs $5 for a model on a timer.

Autodesk–MaintainX release · Viture/NVIDIA release · Meta Wearables toolkit 1.0 · Microsoft Dynamics 365 retirement notice

MARKET · US, PER YEAR, BOTTOM-UP

A $300M wedge inside a $1.8B first market, inside a $9B one



TAM **$9.2B**

15.4M US workers who follow written procedures, at $50 a month for verified records



SAM **$1.8B**

2.47M US maintenance and repair techs: coaching new hires + records for everyone



WEDGE **$306M**

~255k new maintenance techs a year, coached for 6 months at $200 a month

YEAR 3 **$3M ARR**

1% of new techs: ~2,550 trainees, roughly 100 maintenance teams

BLS OEWS national tables, May 2025 · prices are our hypotheses · year 3 is a goal, not a forecast · math in appendix A3

THE OBVIOUS QUESTION

Why not consumers first? Because the budget and the repeat data are in business

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

Autodesk–MaintainX release (28 May 2026) · Frontdoor pricing 2023–2024

THE OTHER OBVIOUS QUESTION

Why can't Meta, Google or Tulip build this? They sell the glasses or the steps. Nobody owns verified work

DEVICES **Meta · Google · Viture**	Sell the glasses and the camera
ASSISTANTS **Meta AI · Gemini**	Answer about one frame. No plan, no state
PRAXIS **Verified work**	Knows the step, checks it, keeps the record
PROCEDURE APPS **MaintainX · Tulip · Augmentir**	Hold the steps. Sign-offs, not sight

IF THEY MOVE

A platform ships a coach: it's our hardware and channel.

Tulip or MaintainX adds cameras: we're their verification layer, or acquisition.

Helix ships: we run on it.

The moat we're building: customer session data, and records nobody wants to rip out.

Public product pages, checked 17 Oct 2026 · Meta Wearables toolkit 1.0 · Tulip Factory Playback announcement

BUSINESS MODEL + GO-TO-MARKET

One saved senior-tech hour a week pays for a seat

$2,400 ÷ 52 ÷ $44 ≈ 1 hour

A $200-A-MONTH SEAT · A YEAR · WEEKS · A SENIOR TECH'S HOURLY COST

FIRST 10

Founder-led paid pilots with facilities teams and service shops

NEXT 100

Case studies, plus integrations with the apps shops already use

THEN 1,000

Distributors and glasses partners; land one site, then every site

OUTREACH: 14 EMAILS SENT 17 OCT · 56 SCHEDULED FOR 19 OCT

Pricing is our hypothesis, anchored on Tulip's $100–250 per station per month · run cost about $1.08 an hour

WHY ME

I've built real-time vision on messy video before, and I built this in a week

LipTrac

Real-time lip reading from ordinary video. 60,000+ frames, 12 testers, provisional patent filed.

7 years

of competitive robotics, coaching teammates without seeing what they see.

1 week

to build the engine, a 48-video eval and a live app, solo. First commit Oct 17.

Monish Saravana · Harvard College · solo founder. Also: path planning for oil-cleanup boats from aerial imagery.

THE LONG GAME

We're building the record of how physical work gets done right

NOW

Coach new technicians

Paid per trainee. Every session is labeled data.

GATE: PAID PILOTS HIT THEIR METRIC

NEXT

Verify and record critical work

The record quality and audit teams pay for.

GATE: CUSTOMERS EXPAND WITHOUT US

LATER

Check any worker or robot

Our data and models train and check robots.

GATE: A PAYING ROBOT PARTNER

Valuations: Figure $39B (Sept 2025) · Physical Intelligence $5.6B (Nov 2025) · Skild AI >$14B (Jan 2026)

THE ASK

$10k makes a first paid maintenance pilot true by April

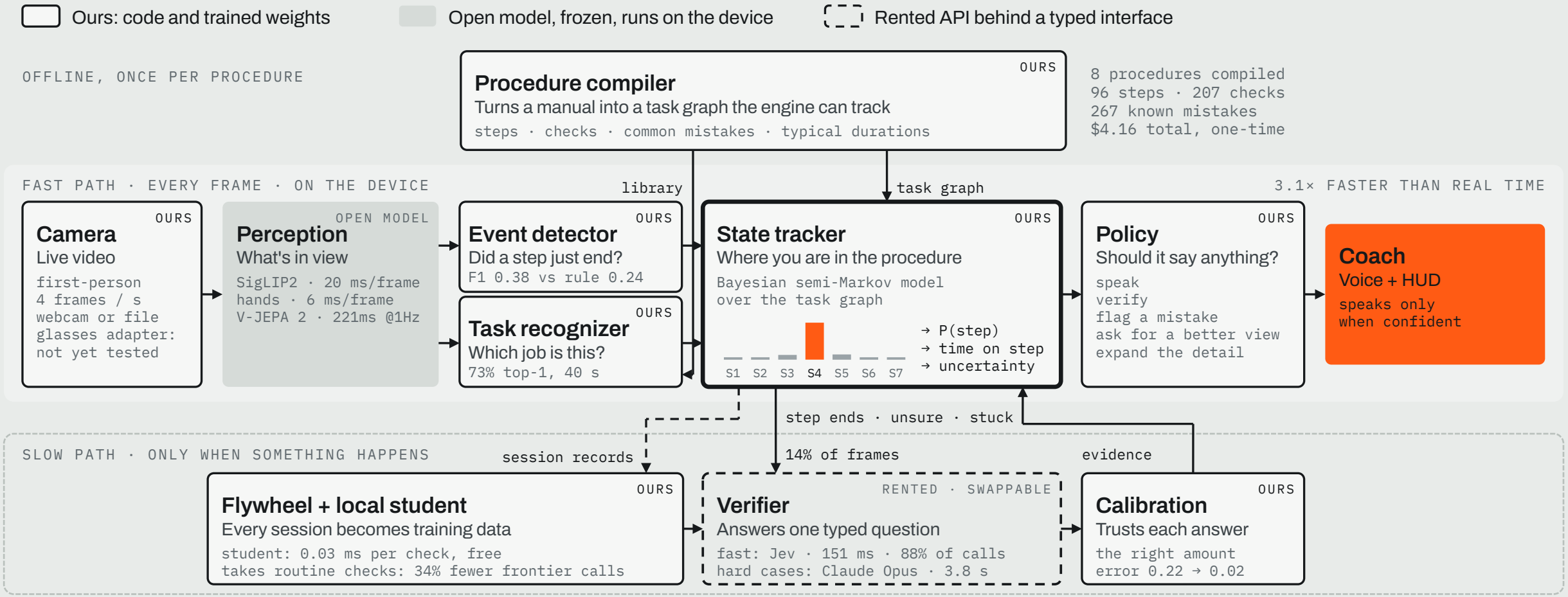
TODAY		BY APRIL
Measured on public cooking video	→	**Measured on real maintenance footage**
Runs live on a laptop	→	**Running on glasses at a real site**
Mistake alerts: 4 of 17 named a real error	→	**Mistake detection re-measured on maintenance**
No customers yet	→	**A first paid pilot, judged on time to work alone**

$3,000 MODELS + GPUS · $2,500 GLASSES + CAPTURE · $2,000 RECORDING MAINTENANCE TASKS · $1,500 INFERENCE · $1,000 FORMATION

monishsaravana@college.harvard.edu

Budget as submitted in the application

The frontier model is one swappable part. The tracking engine around it is ours



Measured on 48 held-out first-person clips from a public dataset (CaptainCook4D)

The status quo: what each tool does today, and what it misses

SHADOWING

A senior tech, side by side

The real teacher today

MISSES About $44 an hour of senior time, one trainee at a time

MANUALS + VIDEOS

PDFs, binders, YouTube

Explain what should happen

MISSES Can't see what you're actually doing

PROCEDURE APPS

MaintainX, Tulip, Augmentir

Put the steps on a phone; collect sign-offs and photos

MISSES A checkbox isn't proof the step was done right

FIXED CAMERAS

Retrocausal, Invisible AI

Watch one workstation on a line

MISSES Maintenance moves around the machine

PRAXIS

Watches the work, follows the person around the machine, and runs on hardware a shop already owns.

Competitor pages checked 2026-10-17

Market math, and the assumptions behind it

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM	1% of new maintenance techs ≈ 2,550 trainees a year, ~100 mid-size teams	2,550 × $200 × 6 months	≈ $3M ARR

Prices are hypotheses. $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

BLS OEWS national tables, May 2025 · BLS Occupational Outlook Handbook

Unit economics, and the pricing risk

DESIGN	$ / VIDEO-MIN	$ / TRAINEE-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

THE RISK

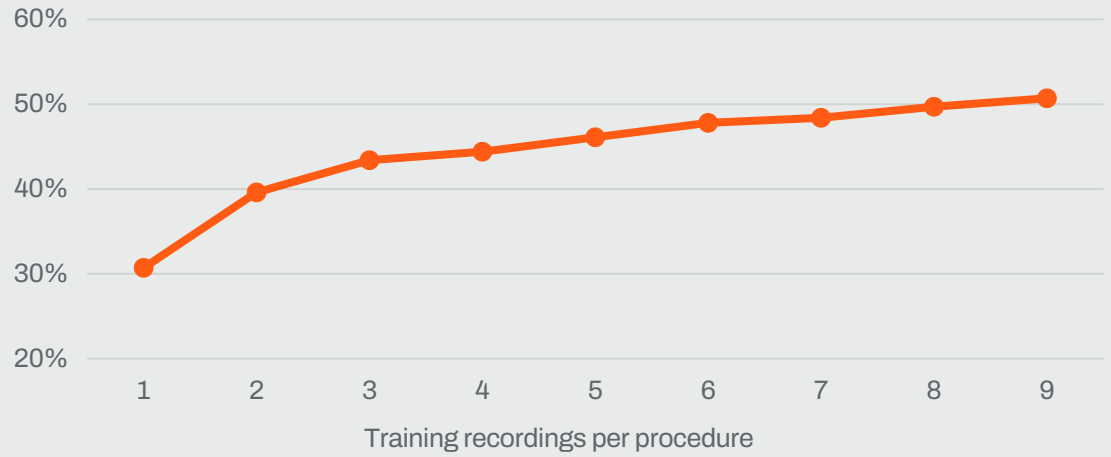
All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

2,520 coached minutes a month = 2 h/day × 21 days · senior-tech cost: BLS

Both learning curves: step models and the local student

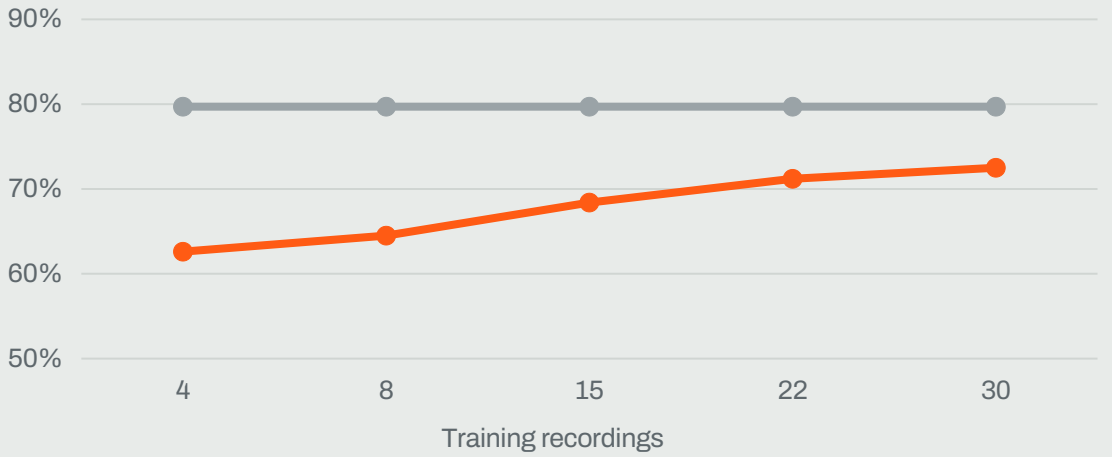
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

checking a step is done · grey: the frontier model, 79.7%



TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

Random subsets of public train-split recordings, 5 seeds per point

Training data, stated plainly

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

Who else is here, and how we differ

CATEGORY	EXAMPLES	US VS THEM
Instruction platforms	Tulip, Augmentir, Dozuki	They show the steps; we watch the work. Natural integration partners
Line cameras	Retrocausal, Invisible AI, Tulip Vision	Fixed at a station on a production line. Can't follow a person around a machine
First-person coaching	Viture Helix + NVIDIA, Strivr	Closest. Hardware-led or high-touch; no published accuracy
Toolkits	NVIDIA XR AI, Meta's camera toolkit, Android XR	Complements: the plumbing we build on
General assistants	Meta AI, Gemini Live	Answer questions about a frame. No plan, no state, no calibration

If a big platform ships a coach: they're building our hardware and plumbing. We go deep in one vertical, with its procedures, records and data.

Public product pages, checked 17 Oct 2026

The risks we see, and what we do about each

RISK	WHAT WE DO
Meta or Google ships a coach	They're building our hardware and plumbing. We go deep in maintenance with its procedures, records and data, and run on their glasses
Glasses aren't ready	It works on a tablet or phone today. Glasses are one more input adapter
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Privacy, unions, monitoring laws	Fast layers run locally; only event keyframes leave (14% of frames). Coaching for the worker, aggregates for the employer. Blur next
A wrong prompt on a safety-critical step	Lockout and similar steps stay the tech's job. Praxis prompts and records them, and never approves them
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the pilot

Glasses and voice: what's built, and what's designed

BUILT

- Runs live in real time on a recorded stream, with the local layers at about 50 ms per frame
- Model calls run in the background, so the live loop never waits (worst lag 0.8 s)
- Speaks its instructions; an urgent line cuts the queue
- Praxis Studio: the engine in a browser, on a camera, an upload or a test clip

DESIGNED, NOT BUILT

- Glasses input adapters (Meta camera toolkit, Android XR), one adapter pattern per device
- Voice input in natural language: "skip that", "why?", "no, I'm still on the belt"
- Push-to-talk by default; a protection headset in loud plants
- Face and badge blur; a local-model tier for sensitive sites

We say "designed to run on" glasses until an adapter has run on that device